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Deep Neural Network-Based Sign Language Translation Model with Dynamic Coordinate Compression

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Received November 6, 2024; revised April 24, 2025; revised May 28, 2025; accepted July 2, 2025; published July 31, 2025

Abstract

This study proposes a deep neural network (DNN)-based Dynamic Sign Language Coordinate Compression (DDSC) model that significantly reduces training time while preserving the semantic integrity of sign language during training and translation. The DDSC model analyzes the Euclidean distances between coordinates extracted from each video frame and selectively compresses segments that exhibit repetitive or minimal motion, while preserving frames with substantial movement that are crucial for conveying meaning. Frame intervals with low inter-frame distance are interpreted as repetitions of the same sign and are compressed to reduce redundancy. In contrast, frames with larger movement variation are retained to prevent semantic loss. The proposed DDSC model was evaluated using a Korean Sign Language (KSL) dataset consisting of word-level video samples for 16 frequently used signs, collected from the AI-Hub platform. The experiment compared three types of coordinate input data: uncompressed, uniformly compressed, and dynamically compressed using DDSC. The corresponding translation accuracies were 0.945, 0.776, and 0.853, respectively. Compared to uniform compression, the proposed DDSC model achieved approximately a 60.73% reduction in training time with only a 9.74% drop in accuracy. These results demonstrate that the DDSC model offers a practical trade-off between accuracy and computational efficiency, making it well-suited for deployment in real-time or resource-constrained environments such as mobile or embedded systems.

Keywords: Deep Neural Network, Dynamic Coordinate Compression, Sign Language Translation

1. Introduction

1.1 Research Background

Although written text may seem like a convenient medium for communication between deaf and hearing individuals, in practice, 27% of deaf individuals have difficulty understanding Korean, and an additional 43% only partially understand it. As a result, while many hearing individuals assume that deaf people use Korean to express their thoughts, a significant proportion (43%) consider Korean a foreign language and do not fully comprehend it [1].

Sign language translation is thus an essential tool for enabling individuals with hearing impairments to participate more fully in society. As of 2022, the average number of sign language interpretation centers per region in Korea is 12.7, while the number of individuals with severe hearing impairments is approximately 5,217. This results in a ratio of approximately 102.4 individuals per interpreter [2]. Due to this shortage, individuals with hearing impairments often face difficulties in clearly expressing their intentions, which restricts their access to employment opportunities and essential public services such as healthcare and legal support. Consequently, they are unable to receive services equal to those available to hearing individuals [2].

Given this critical shortage of professional sign language interpreters, deep learning-based sign language translation systems offer a scalable and sustainable alternative for bridging communication gaps. However, training such models on large-scale sign language datasets typically requires significant computational resources and time, limiting their real-world applicability—especially in mobile or embedded environments. To address this limitation, the proposed DDSC model dynamically compresses coordinate sequences by removing redundant frames while preserving semantically important gestures. This approach significantly reduces training time while maintaining translation accuracy, thereby enabling more efficient and practical deployment.

1.2 Previous Research

Research on translating Korean sign language (KSL) using existing deep learning technologies has been actively conducted using various architectures. Several studies employed convolutional neural networks (CNN) [3-6], while others explored long short-term memory (LSTM) models [7-8] or combined CNN and LSTM for enhanced performance [5, 9]. A CorrNet-based model integrating CNN and BiLSTM was introduced in [10] to learn frame-wise correlations in continuous sign language recognition. Transformer-based models were adopted in [11-12] to implement end-to-end translation frameworks and fine-grained motion context learning, respectively. Graph convolutional networks (GCNs) were applied in [13] to enhance skeleton-based action recognition through position-aware spatio-temporal modeling. Additionally, research conducts the integration of Google's Inception v3 model with the Support Vector Machine (SVM) machine learning technique [14]. **Table 1** summarizes the deep learning technologies, criteria, and the usage of coordinate extraction and compression employed in previous sign language translation studies.

Table 1. Previous deep learning-based research on sign language translation

Studies	Models	Criteria	Coordinate Extraction	Coordinate Compression
Kim et al. [3]	CNN	Accuracy	○	×
Kim et al. [4]	CNN	Accuracy	○	×
Gil et al. [5]	CNN, LSTM	Accuracy	○	×
Lee et al. [6]	YOLOv7	mAP	×	×
Lee et al. [7]	LSTM	Accuracy	○	×
Kim et al. [8]	LSTM	Accuracy	○	×
Kim et al. [9]	CNN, LSTM	Accuracy	○	×
Hu et al. [10]	CNN, BiLSTM	Accuracy	○	×
Camgoz et al. [11]	Transformer	BLEU, Accuracy	○	×
Lee et al. [12]	Transformer	Accuracy	○	×
Yang et al. [13]	GCN	Accuracy	○	×
Abiyev et al. [14]	Inception v3, SVM	Accuracy	×	×
This work	DNN	Accuracy	○	○

According to **Table 1**, although research has been conducted on translating sign language using deep learning and evaluating accuracy, no studies have yet been undertaken to compare performance through the compression of sign language coordinates. Therefore, this study evaluates the performance of the proposed deep neural network (DNN)-based dynamic sign language coordinate compression (DDSC) model by comparing the accuracy and training time of sign language translation using DNN with original coordinates, uniformly compressed coordinates, and coordinates dynamically compressed using the proposed DDSC model.

As shown in **Table 1**, coordinate compression remains a unique contribution of the proposed DDSC model, offering practical advantages for real-time and resource-constrained applications. Furthermore, in real-world environments such as mobile or embedded sign language translation systems, computational resources are limited compared to high-performance desktop systems. Processing raw sign language coordinate data under such constraints can be both inefficient and time-consuming. To address this, the DDSC model dynamically compresses redundant coordinate frames based on motion variation between adjacent frames, while preserving semantically meaningful gestures. This approach not only enhances computational efficiency but also retains the essential features necessary for accurate translation.

This paper is organized as follows. Section 2 describes the proposed DDSC model and outlines the process for evaluating the model's performance. Section 3 presents the simulation results of the proposed model, providing a comparative analysis of the outcomes for the three scenarios: original coordinates, uniformly compressed coordinates, and dynamically compressed coordinates. Finally, Section 4 summarizes the conclusions.

2. Methodology

The proposed DDSC model is designed to preserve the meaning of sign language as much as possible during the compression of its coordinates while significantly reducing the time required for a DNN to train on those coordinates. To maintain the accuracy of the sign language's meaning, the proposed DDSC model calculates the Euclidean distance between the coordinates extracted from each frame and those from the subsequent frame. Frame intervals

with shorter distances are considered segments where the same sign language action is repeated, and these are compressed to reduce the number of frames. In contrast, intervals with larger distances indicate substantial hand movements that change the meaning of the sign language; therefore, these frames are preserved to ensure that the meaning is not compromised.

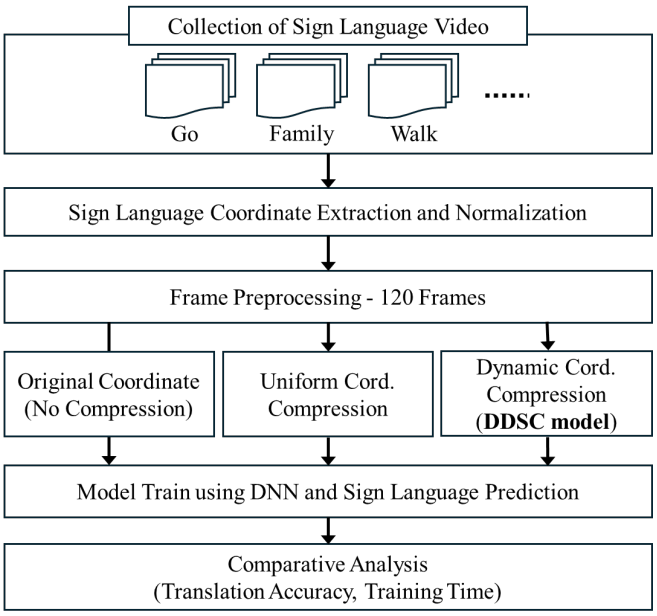


Fig. 1. Workflow for evaluating translation accuracy and training time across different coordinate compression models including the proposed DDSC model

Fig. 1 illustrates the evaluation process for comparing translation accuracy and training time when the DNN learns and translates sign language coordinates.

This study evaluates the performance of the proposed DDSC model by comparing and analyzing sign language translation accuracy and training time when training and translating three coordinate scenarios using a DNN. The first scenario, referred to as 'original coordinates,' comprises sign language coordinates extracted from sign language videos without compression. The second scenario, termed 'uniform coordinates,' involves compressing the sign language coordinates into uniform frame units. The third scenario, known as 'dynamic coordinates,' employs the proposed DDSC model to compress the coordinates while preserving the meaning of the sign language as much as possible. These three coordinate scenarios are trained using a DNN, and their accuracy and training time during sign language translation are compared and analyzed to assess the performance of the proposed DDSC model.

The following section provides a detailed explanation of the sequential procedure for comparing and analyzing the performance of the three scenarios.

2.1 Collection of Sign Language Video

This study utilizes KSL videos provided by the National Information Society Agency's AI-HUB [15], which offers a total of 536,000 sign language videos. All datasets used in this study have the same resolution of 1920×1080, ensuring consistency in the input video quality during coordinate extraction and preprocessing. For this research, 16 representative sign language words commonly used in daily life are selected to compare and analyze the sign language translation accuracy and training times of the three scenarios. In this study, a total of 16

frequently used KSL words were selected based on the availability of sufficient high-quality samples from the AI-Hub dataset. This controlled experimental setup was designed to enable consistent evaluation of the proposed DDSC model under various coordinate compression scenarios, while minimizing the impact of confounding factors such as class imbalance and data sparsity.

Table 2. Number of video samples for each selected KSL word

Word	Count	Word	Count	Word	Count	Word	Count
Walk	261	Borrow	144	Help	141	Recover	96
Blocked	192	Give	144	Disappear	137	Difficult	93
Brazen	189	Tolerate	144	Consideration	135	Good	93
Go	186	Correct	141	Sleep	132	Family	48

Table 2 presents the 16 sign language words used in this study, along with the number of video samples available for each word. These words were selected based on their frequent use in daily life and the availability of sufficient high-quality data, ensuring consistency and fairness in evaluating the translation accuracy and training time across the three coordinate compression scenarios. A total of 2,276 sign language video samples were used in this study. The dataset was divided into training and validation sets in an 80:20 ratio, with class-balanced sampling applied to ensure uniform representation of each word across both subsets.

2.2 Sign Language Coordinator Extraction and Normalization

In this study, the MediaPipe library is utilized to extract sign language coordinates from sign language videos. MediaPipe, developed by Google, is an open-source library capable of performing tasks such as real-time action recognition, object detection, face detection and tracking, and hand gesture recognition [16].

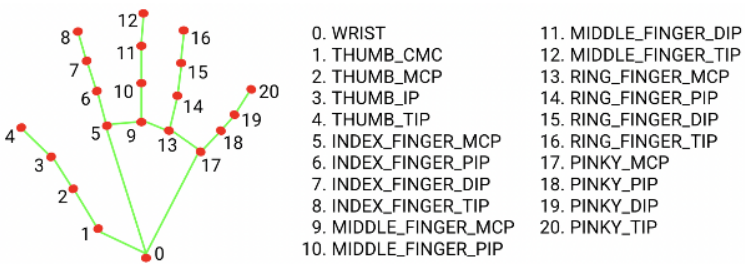
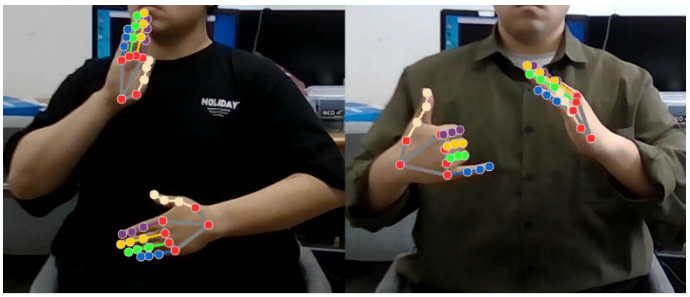


Fig. 2. The 21 hand keypoint landmarks extracted by MediaPipe [16]

Fig. 2 illustrates the 21 landmarks associated with the wrist, thumb, index finger, middle finger, ring finger, and little finger that can be detected by MediaPipe. Each landmark is extracted as a three-dimensional coordinate represented by x, y, and z values.

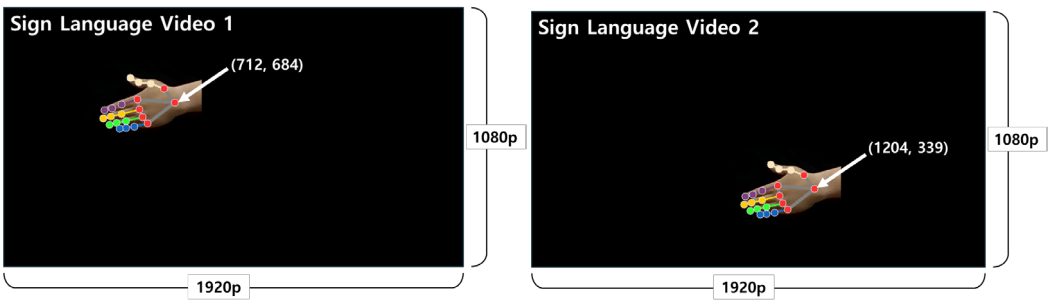


(a) (b)

Fig. 3. Hand landmark coordinates of KSL gestures: (a) ‘correct’, (b) ‘family’

Fig. 3 presents examples of sign language landmarks extracted from the 'correct' and 'family' movements in sign language videos. The coordinates, extracted using MediaPipe, consist of 42 landmarks (21 per hand) with three-dimensional x, y, and z coordinates for each. Each sign language video is preprocessed into 120 frames, resulting in sign language coordinate data of size (120, 42, 3). The following subsection provides a detailed explanation of the frame preprocessing.

Additionally, the sign language coordinates extracted through MediaPipe may yield different coordinate values for the same sign language, depending on the position of the hands on the screen, which is influenced by the size of the input image. To address this issue, the extracted coordinates are normalized to values between 0 and 1 to ensure consistency in the sign language coordinates.



(a) (b)

Fig. 4. Two sign language videos expressing the same gesture but with different spatial positions in 1920×1080 resolution

Fig. 4 (a) and **(b)** depict two distinct sign language videos that convey the same meaning. Although both figures represent the same sign language meaning, their coordinate values differ—(712, 684) for **Fig. 4 (a)** and (1204, 339) for **Fig. 4 (b)**—which undermines the consistency of the sign language representation. Therefore, this study scales 21 coordinates to values between 0 and 1, as illustrated in **Fig. 4**, to ensure the consistency of the sign language coordinates.

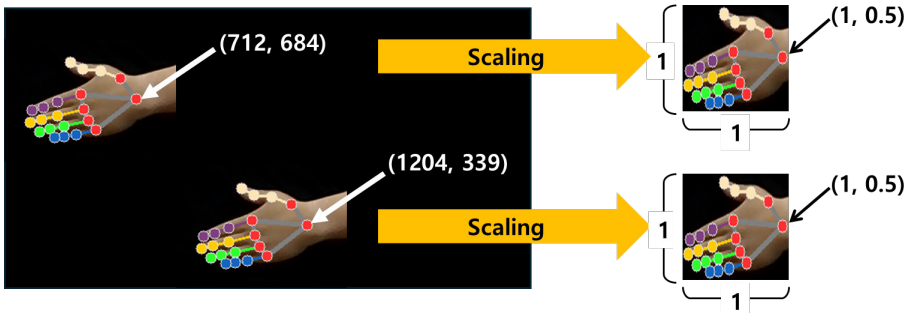
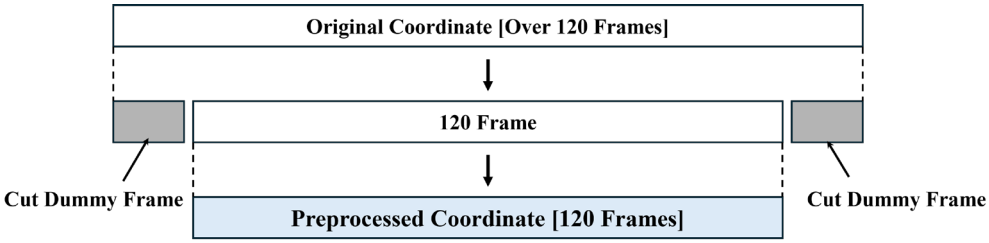


Fig. 5. Normalization of sign language coordinates through scaling to a consistent reference frame

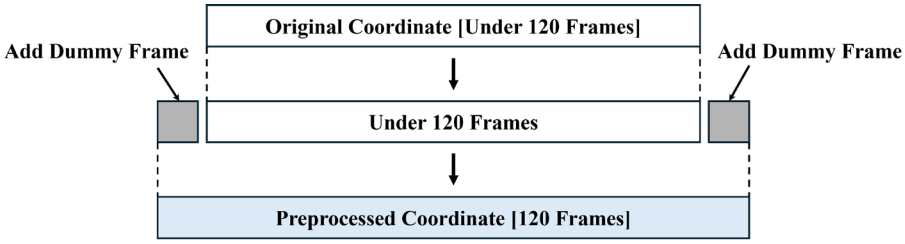
Fig. 5 illustrates the process of transforming coordinates from two different sign language videos that convey the same meaning into uniform coordinates through scaling, thereby ensuring consistency in the representation of the same sign language meaning from the two distinct sets of coordinates.

2.3 Frame Preprocessing

Since the lengths of the collected sign language videos vary, a preprocessing step was applied to standardize all videos to 120 frames by adding or removing stationary frames, ensuring that the semantic content of the sign language is preserved. In this study, videos with a maximum length of approximately 130 frames were selected. In most cases, the initial and final ~10 frames of such videos captured the signer in a static posture without performing any meaningful gestures. Therefore, these segments were considered non-semantic dummy frames, as the extracted coordinate values did not contribute to conveying the intended sign language meaning. These frames were removed to normalize the sequence to 120 frames. Conversely, for videos shorter than 120 frames, an equal number of static frames were appended to the beginning and end to maintain semantic integrity while achieving a consistent sequence length. For example, when a video exceeds 120 frames, non-informative static frames are trimmed symmetrically from both ends. When a video is shorter than 120 frames, an equal number of static frames are added to both ends, resulting in a uniform input length across all samples.



(a)



(b)

Fig. 6. Preprocessing of sign language coordinate frame (a) Trimming dummy frames for sequences exceeding 120 frames. (b) Padding dummy frames for sequences shorter than 120 frames

Fig. 6 illustrates the preprocessing procedure for sign language coordinate frames. In **Fig. 6 (a)**, when the original sign language coordinates exceed 120 frames, static frames without motion at the beginning and end of the video are removed. **Fig. 6 (b)** demonstrates the addition of equal numbers of static frames to the start and end of the video when the original coordinates are shorter than 120 frames, thereby standardizing the length of all sign language coordinate frames to 120 frames.

The sign language videos standardized to 120 frames are compressed to 30 frames using both the existing uniform compression technique and the proposed DDSC model. The uniform compression technique does not consider the relationships between sign language coordinates across frames and compresses the data uniformly at fixed intervals [17]. To fairly evaluate the performance differences between the proposed dynamic compression technique and the existing uniform compression technique, all sign language videos are designed to have their frame counts adjusted to 30 frames after compression. Subsequently, the sign language coordinate data compressed to 30 frames are utilized for training through a DNN for sign language translation.

2.4 Compression of Sign Language Coordinate

2.4.1 Uniformly Compressed Coordinate

This subsection describes a technique for uniformly compressing coordinates extracted from each frame of sign language videos without considering the interconnectivity of the coordinates [17]. The following **Fig. 7** illustrates an example of compressing the original 120 frames into 30 frames by calculating the average coordinates from every four frames and consolidating them into a single frame.

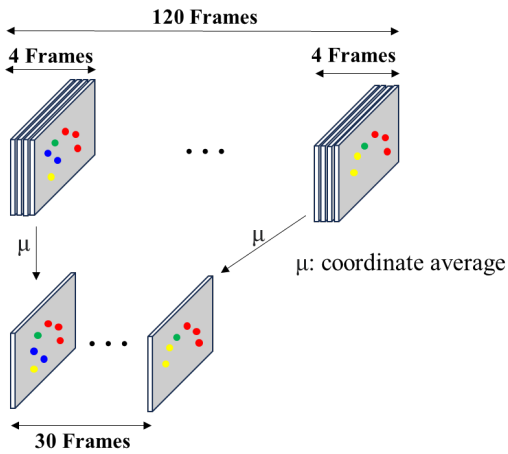


Fig. 7. Illustration of the uniform coordinate compression model

Fig. 7 illustrates the technique of uniformly compressing sign language coordinates extracted from every four frames into a single frame within a sign language video composed of a total of 120 frames. The previous uniform compression technique compresses the frames by calculating the average of the coordinates from uniform frame units, thereby neglecting the interrelationships between the sign language coordinates of adjacent frames.

The following subsection describes the technique of dynamically compressing sign language coordinates using the proposed DDSC model. This model takes into account the interconnectivity of sign language coordinates between frames to preserve the meaning of the

sign language as much as possible, rather than employing uniform compression.

2.4.2 Dynamically Compressed Coordinate by proposed DDSC model

This subsection explains the technique of dynamically compressing coordinates using the proposed DDSC model, which takes into account the interrelationships between coordinates across frames. The uniform coordinate compression technique described in Section 2.4.1 compresses the coordinates simply into uniform frame units without considering the interrelationships between the coordinates across frames. In contrast, the dynamic coordinate algorithm proposed in this study calculates the interrelationships of the coordinates extracted from each frame using Euclidean distance.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \tag{1}$$

where, (x_1, y_1, z_1) and (x_2, y_2, z_2) represent the x, y, and z coordinates of the sign language coordinates for the current frame and the next frame, respectively.

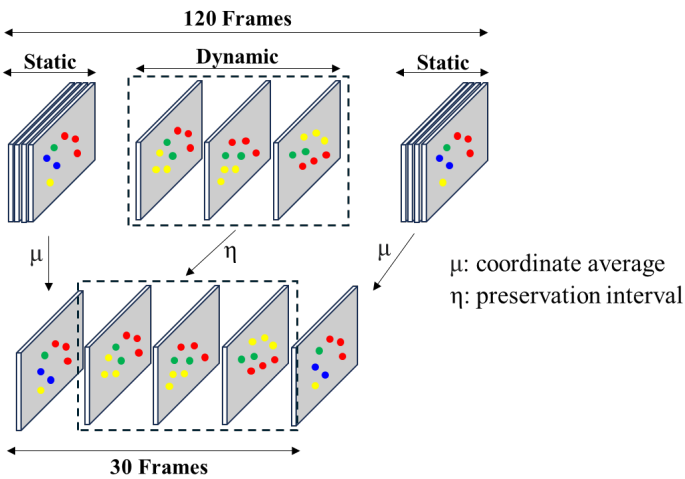


Fig. 8. Illustration of the dynamic coordinate compression using the proposed DDSC model

Fig. 8 illustrates the concept of dynamically compressed coordinates using the proposed DDSC model. The circles in **Fig. 8** represent the sign language coordinates. Each of the 42 landmarks (21 per hand × 2 hands) is analyzed independently to calculate its movement. The overall magnitude of movement is then determined based on the average motion across all 42 coordinates, and frames are compressed accordingly. The proposed DDSC model calculates the distance between sign language coordinates across consecutive frames. Frame intervals with shorter distances are considered as segments where the same sign language gesture is repeated and are thus compressed as much as possible to reduce the number of frames. Conversely, frame intervals with larger distances reflect significant hand movements that convey the meaning of the sign language. These frames are preserved to ensure that the meaning of the sign language is maintained.

2.5 Comparison of Uniform and Dynamic Compression Techniques

This subsection describes the differences in compressing a sign language video composed of 120 frames to 30 frames using conventional uniform compression technique and the dynamic coordinate compression technique by proposed DDSC model.

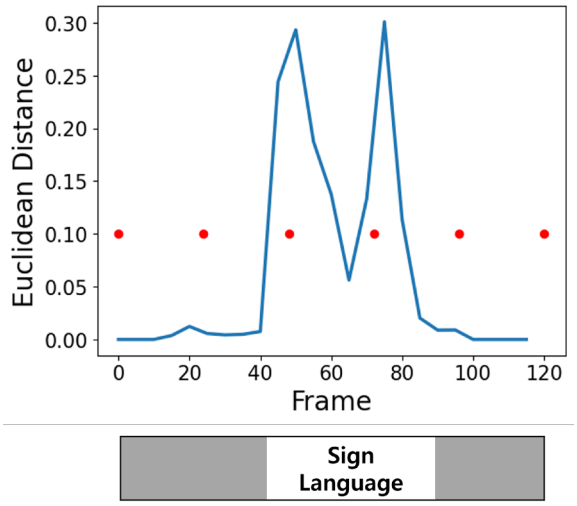


Fig. 9. Euclidean distance between coordinates and selected compression intervals

Fig. 9 presents the distances between sign language coordinates across frames, calculated using Equation (1). The blue line represents these distances, while the red circles indicate the intervals of frames that can be compressed. The uniform compression technique does not account for the distances between adjacent frames or the interconnectivity of the coordinates; thus, it can be observed that the distances represented by the red circles are identical.

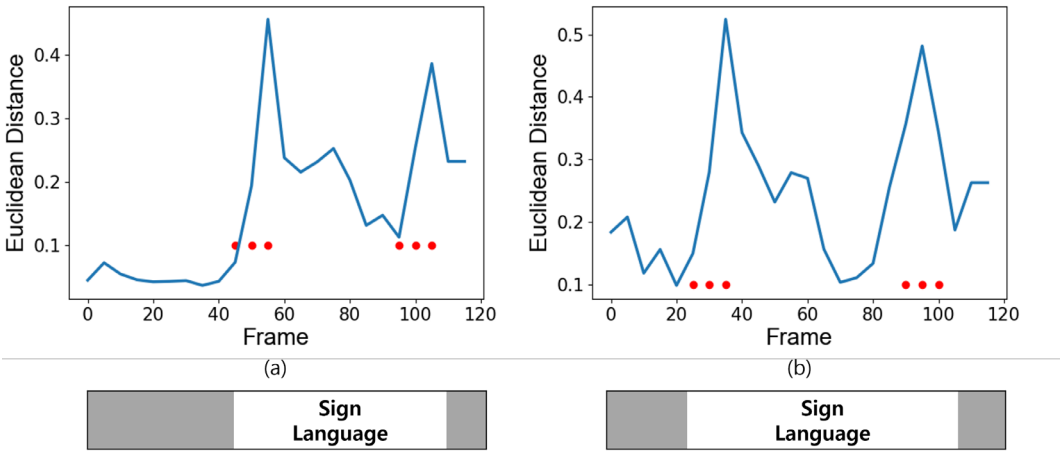


Fig. 10. Euclidean distance trends and dynamically selected compression intervals for sign language sequences

Fig. 10 illustrates an example of the distance between frames when employing dynamic compression techniques on two sign language videos depicting "walking." The red circles indicate intervals where compression occurs. According to **Fig. 10**, these red circles are

marked in regions where there is a significant distance between the sign language coordinates, specifically indicating intervals where the signer's hands move considerably. This denotes frames where compression should be minimized to preserve the meaning of the sign language. Conversely, the intervals marked with gray bars represent intervals where the distance between sign language coordinates is short, indicating regions where similar sign language coordinates appear repeatedly, thereby signifying frames that can be compressed as much as possible.

As each sign language video may vary in motion dynamics, signer posture, camera resolution, and the distance between the signer and the camera, methods relying on a fixed threshold often struggle to generalize effectively. In contrast, the proposed DDSC model does not require a predefined threshold; instead, it dynamically selects key frames based on the point of highest coordinate variation within each video. This adaptive process enables the model to operate effectively across diverse video inputs without manual hyperparameter tuning.

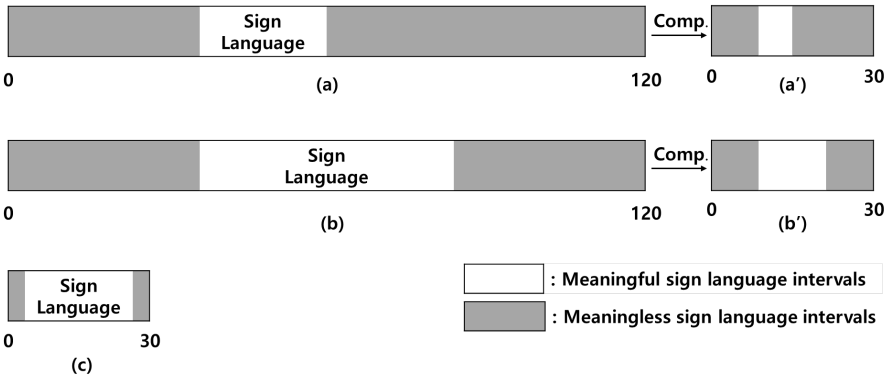


Fig. 11. (a), (b) Original coordinates of "walking" from different sign language videos; (a′), (b′) uniformly compressed versions; (c) dynamically compressed coordinates using the proposed DDSC model.

Fig. 11 illustrates the bars that distinguish the compression intervals of sign language video frames, highlighting the differences between uniformly compressed coordinates and those dynamically compressed using the proposed DDSC model, while also emphasizing the advantages of preserving sign language coordinates.

Fig. 11 visualizes the results of compressing two identical sign language video clips using both conventional uniform compression techniques and the proposed dynamic compression techniques. The gray intervals indicate areas where the distance between coordinates is short due to the proximity of adjacent frames, representing states that do not convey the meaning of the sign language, such as standing or waiting poses. These intervals signify regions where frames can be compressed as much as possible. In contrast, the white intervals correspond to regions where the signer's hands move significantly, indicating areas where the meaning of the sign language changes; therefore, these frames should be preserved to the greatest extent possible.

For instance, **Fig. 11 (a)** and **(b)** both depict the sign for "walking," where **Fig. 11 (a)** shows a faster gesture expression, and **Fig. 11 (b)** represents a slower one. **Fig. 11 (a′)** and **(b′)** illustrate the results of applying uniform compression to **Fig. 11 (a)** and **Fig. 11 (b)**, respectively. However, although both videos represent the same sign language word, the positions of the coordinates in the uniformly compressed versions to **Fig. 11 (a′)** and **Fig. (b′)** differ significantly due to variations in gesture speed. This inconsistency increases the

likelihood of the DNN misclassifying them as different words during training.

In contrast, **Fig. 11 (c)** shows the results of applying the proposed DDSC model for dynamic coordinate compression. This technique selectively preserves the frame intervals that convey the semantic meaning of the sign, while compressing less meaningful segments. By maintaining key gesture information, the DDSC model ensures that the compressed sequences retain their semantic identity, thereby improving the DNN's training accuracy and translation performance.

2.6 Training of Coordinates and Sign Language Translation using DNN

This study employed DNN technology to train and translate sign language, aiming to compare and evaluate the translation accuracy and training speed of the original coordinates, uniformly compressed coordinates, and dynamically compressed sign language coordinates using the proposed DDSC model.

A DNN is a specific type of artificial neural network (ANN) that consists of multiple layers. ANN are developed based on inspiration from biological neurons. The neurons in an artificial neural network generate a single output by multiplying multiple input values by their respective weights and summing the results. Generally, an ANN refers to a neural network with one hidden layer, whereas a DNN refers to a neural network with two or more hidden layers [18].

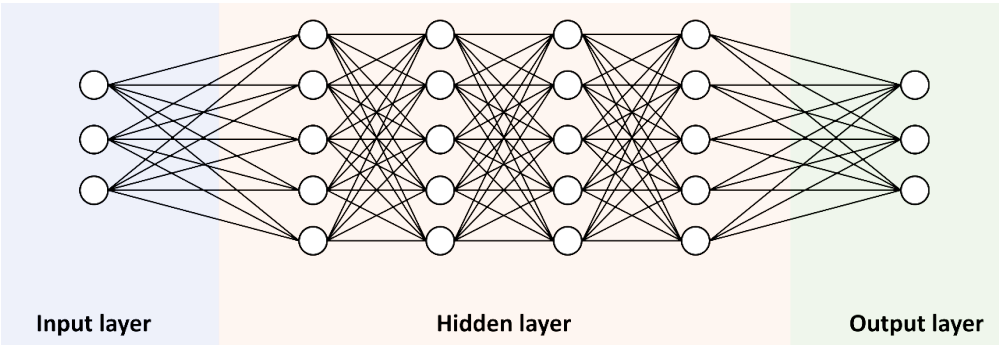


Fig. 12. Architecture of the DNN used in this study

Fig. 12 illustrates the structure of the DNN. This network is characterized by deeply interconnected neurons across multiple hidden layers, enabling it to effectively solve complex problems and extract various features. Nonlinear transformations are performed on the input data within the hidden layers, while the final output layer generates the ultimate results. These processes can be described mathematically, and the operations of the hidden and output layers can be represented by the following equations.

$$h_j = a\left(w_{i,j} \cdot x_i + b_j\right) \tag{2}$$

Where h_j, a, $w_{i,j}$, x_i, and b_j represent the output node, activation function, weight, input node, and bias, respectively. Each hidden layer utilizes the output from the previous layer as its input and transmits the output to the subsequent layer. An activation function may be applied prior to transmitting the output. The output of the final hidden layer, after traversing all hidden layers, signifies the prediction result.

In this study, to learn and translate sign language coordinates using a DNN, the coordinates of the sign language, compressed to 30 frames, are preprocessed into a data format of 1×3,780 and subsequently fed into the input layer. Afterwards, the DNN model is trained on the preprocessed data and predicts one of the 16 sign language words in the output layer.

Table 3. Input feature structure and dimensionality of the three coordinate models

Model	No. of Landmarks	No. of spatial coordinates	No. of Frames	No. of Input Layers
Original Coord.	42	3	120	15,120
Uni. Comp. Coord.	42	3	30	3,780
Dyna. Comp. Coord.	42	3	30	3,780

Table 3 summarizes the structure of the input data across the three scenarios. Each sign language frame consists of 42 landmarks, each with 3 spatial coordinates (x, y, z), resulting in 126 features per frame. For original coordinates using 120 frames, the total number of input features is 15,120 (42 × 3 × 120). In contrast, both the uniformly and dynamically compressed coordinates use 30 frames, reducing the input size to 3,780.

3. Simulation Results

In this section, DNN are employed to evaluate the performance of the proposed deep learning-based dynamic coordinate compression technique by comparing and analyzing the sign language translation accuracy and training time across the following three scenarios:
- Scenario 1: Original coordinates (120 frames)
- Scenario 2: Uniformly compressed coordinates (30 frames)
- Scenario 3: Dynamically compressed coordinates (30 frames, DDSC model)

3.1 Simulation Environment

Table 4 illustrates the experimental simulation environment established for the comparative analysis of sign language translation accuracy and training time across the three scenarios: original coordinates, uniformly compressed coordinates, and dynamically compressed coordinates using the proposed DDSC model. All experiments were conducted on a workstation equipped with an Intel(R) Xeon(R) CPU @ 2.00GHz, 16GB of RAM, and an NVIDIA Tesla P100 GPU. The models were implemented and trained using Python 3.10.13 and PyTorch 2.1.2.

Table 4. Experimental simulation environment

Parameter	Specification
CPU	Intel(R) Xeon(R) CPU @ 2.00GHz
GPU	NVIDIA Tesla P100
Memory	16GB
Simulation Tools	Python 3.10.13 / PyTorch 2.1.2

Furthermore, **Table 5** presents the simulation parameters.

Table 5. Simulation parameters

Parameter	Original Coord.	Uni. Comp. Coord.	Dyna. Comp. Coord.
No. of input layer	15,120	3,780	3,780
No. of hidden layer 1	7,560	1,890	1,890
No. of hidden layer 2	3,780	945	945
No. of output layer	16	16	16
No. of epochs	300		
Initial learning rate	1e-6		
Optimizer	Adam [19]		

The number of input units, hidden units, and training epochs were configured as shown in **Table 5** to enable a fair comparison of translation accuracy and training time across the three scenarios: original coordinates, uniformly compressed coordinates, and dynamically compressed coordinates. The input size differs across scenarios due to variations in the number of frames, while the output layer size remains consistent at 16, corresponding to the number of target classes. All models were trained for 300 epochs with an initial learning rate of 1e-6. The Adam optimizer was employed to improve convergence stability and avoid issues related to local minima [19].

3.2 Comparative Analysis of Sign Language Translation Accuracy

This section presents the results of a comparative analysis of sign language translation accuracy and training time across the three scenarios: original coordinates, uniformly compressed coordinates, and dynamically compressed coordinates.

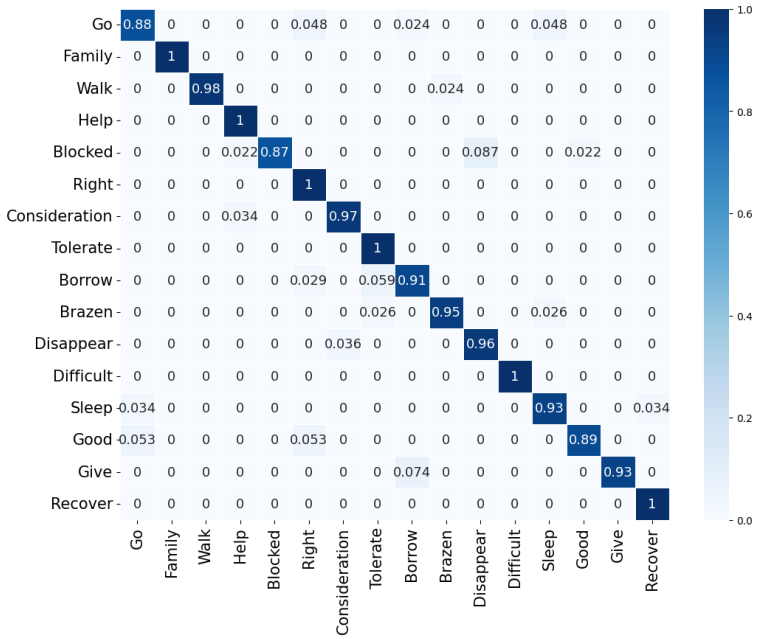


Fig. 13. Confusion matrix of sign language translation accuracy for the original coordinates

Fig. 13 illustrates the confusion matrix for the sign language translation accuracy of the uncompressed original coordinates. The results presented in **Fig. 13** indicate that the average translation accuracy for the 16 sign language words is 0.945, which confirms that it is higher than the translation accuracy achieved when training and translating with both the uniformly compressed coordinates and the dynamically compressed coordinates.

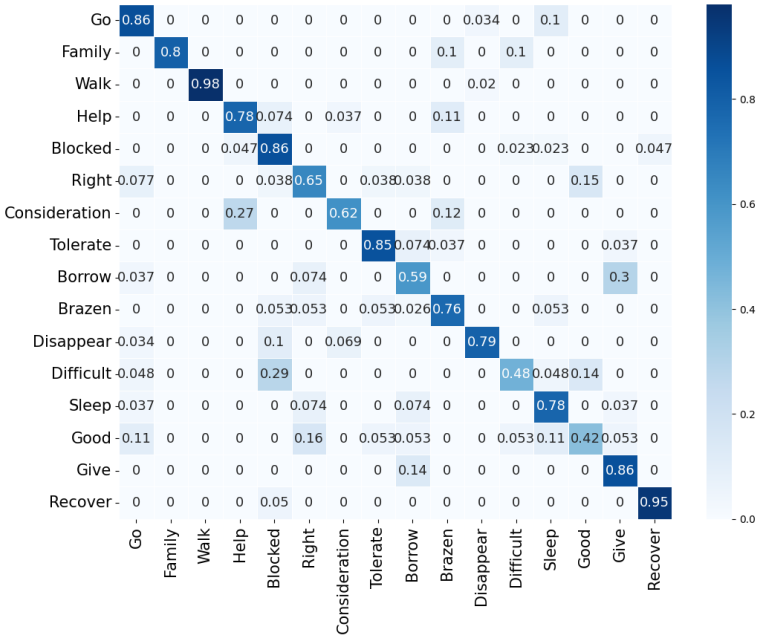


Fig. 14. Confusion matrix of sign language translation accuracy for uniformly compressed coordinates

Fig. 14 illustrates the confusion matrix for the sign language translation accuracy of the uniformly compressed coordinates. The average translation accuracy for all 16 sign language words is 0.776, indicating that this is the lowest accuracy among the three scenarios. This reduction in accuracy can be attributed to the process of uniformly compressing the coordinates without considering the interrelationships between them, which results in a loss of meaning in the sign language and, consequently, a decrease in translation accuracy.

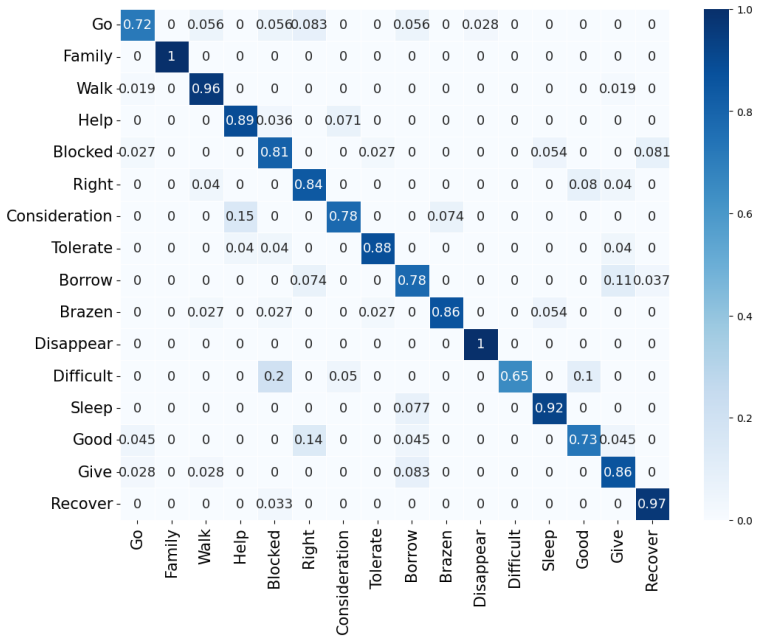


Fig. 15. Confusion matrix for sign language translation accuracy of dynamically compressed coordinates

Fig. 15 illustrates the confusion matrix for the sign language translation accuracy of dynamically compressed coordinates using the proposed DDSC model. According to the simulation results, the overall average accuracy is 0.853, which is lower than that of the original coordinate but represents an increase of approximately 10% compared to the accuracy of uniformly compressed coordinates. This improvement can be attributed to the consideration of interconnectivity among sign language coordinates across frames for efficient compression. Specifically, coordinates for frames with similar movements are compressed, while those for frames in which the signer's hands are in motion are preserved to the greatest extent possible to prevent any degradation of the meaning of the sign language. Consequently, this approach leads to an increase in translation accuracy compared to uniformly compressed coordinates.

As for the drop in accuracy, it primarily results from information loss that occurs during the compression process. However, the proposed DDSC minimizes this loss by selectively preserving frames with significant movements, in contrast to uniform compression that treats all frames equally. This approach enables DDSC to retain more semantic information and achieve relatively higher translation accuracy.

In Subsection 3.3, a comparative analysis of sign language translation accuracy and training times across the three scenarios will be presented.

3.3 Comparative Analysis of Accuracy and Training Time

This subsection compares and analyzes the sign language translation accuracy and training times across three scenarios: original coordinates, uniformly compressed coordinates, and dynamically compressed coordinates using the proposed DDSC model.

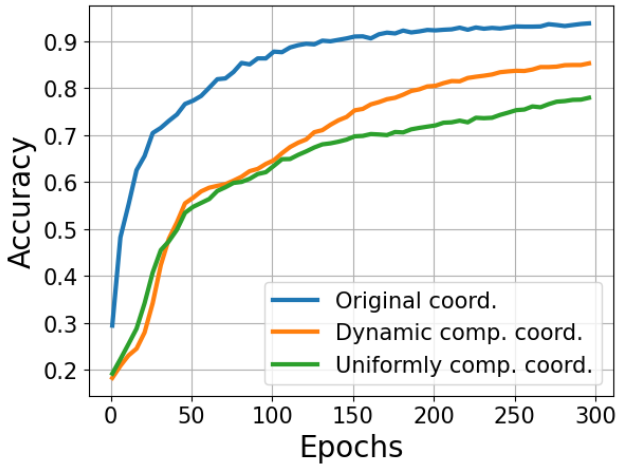


Fig. 16. Accuracy over epochs for original, uniformly compressed, and dynamically compressed coordinate scenarios

Fig. 16 illustrates the changes in sign language translation accuracy as the number of epochs increases for these scenarios. According to **Fig. 16**, as the epoch value increases from 1 to 300, sign language translation accuracy improves for all three scenarios. Furthermore, observations indicate that the highest translation accuracy is achieved with the original coordinates, followed by the proposed dynamic coordinate compression technique and the uniform coordinate compression technique.

Table 6. Accuracy (%) and training time (s) comparison across three coordinate scenarios over epochs

Model	Epoch									
	60		120		180		240		300	
	Acc.	Train. Time	Acc.	Train. Time	Acc.	Train. Time	Acc.	Train. Time	Acc.	Train. Time
Original Coord.	0.789	560.32	0.890	1270.99	0.917	1849.41	0.936	2306.26	0.945	2795.74
Uni. Comp. Coord.	0.559	219.07	0.664	430.73	0.708	646.78	0.735	862.27	0.776	1072.65
Dyn. Comp. Coord.	0.577	219.41	0.684	438.91	0.783	657.22	0.831	873.43	0.853	1097.99

According to **Table 6**, the translation accuracy is highest for the original coordinates across all epoch intervals, with the final accuracy at epoch=300 ordered as follows: original coordinates, dynamically compressed coordinates, and uniformly compressed coordinates. Additionally, in all scenarios, training time increases with the number of epochs. The results also show that the training times for the uniformly and dynamically compressed coordinates differ by approximately 2%, with almost no notable difference between them. Furthermore, at epoch=300, the training time for the original coordinates increased by approximately 160.64% compared to the compressed coordinates.

The difference in training time is attributed to the variation in the input data size across the three scenarios. While the original coordinates are trained using 120 frames of sign language coordinates, the uniformly and dynamically compressed coordinates are reduced to 30 frames, leading to a shorter training time. In the following subsection, the accuracy and training time for learning and translating sign language coordinates across the three scenarios using a DNN

are compared to evaluate the efficiency of each scenario.

3.4 Comparative Analysis of Translation Accuracy and Efficiency

This section compares and evaluates the sign language translation accuracy and training time across the three scenarios to assess the efficiency of translating sign language.

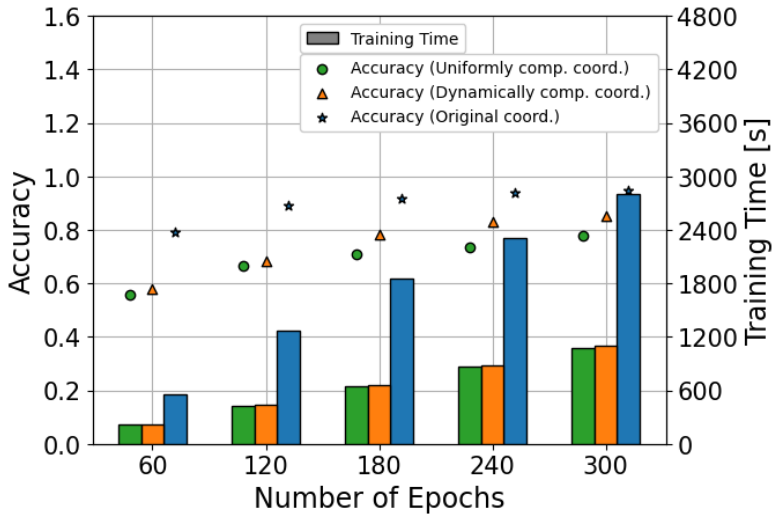


Fig. 17. Comparative analysis of translation accuracy and training time across the three coordinate scenarios

The graph in **Fig. 17** visualizes the data presented in **Table 6**. It illustrates the results of sign language translation accuracy and training time for the three scenarios as the number of epochs changes. The bar graph denotes the training time, while the circles, triangles, and stars represent the translation accuracy for uniformly compressed coordinates, dynamically compressed coordinates, and original coordinates, respectively. According to **Fig. 17**, as the number of epochs increases from 60 to 300, both accuracy and training time increase across all three scenarios.

Table 7. Relative changes in accuracy (%) and training time (s) compared to the original coordinate scenario

Model	Epoch									
	60		120		180		240		300	
	Acc.	Train. Time	Acc.	Train. Time	Acc.	Train. Time	Acc.	Train. Time	Acc.	Train. Time
Uni. Comp. Coord.	-29.15	-60.90	-25.39	-66.11	-22.79	-65.03	-21.47	-62.61	-17.88	-61.63
Dyn. Comp. Coord.	-26.87	-60.84	-23.15	-65.47	-14.61	-64.46	-11.22	-62.12	-9.74	-60.73

Table 7 presents the relative changes in translation accuracy and training time for uniformly and dynamically compressed coordinates, compared to the original coordinates, which achieved the highest average performance. The results indicate that both compression methods exhibit reduced accuracy compared to the original coordinates; however, the gap narrows as the number of epochs increases. At epoch 60, the translation accuracy dropped by

29.15% and 26.87% for uniformly and dynamically compressed coordinates, respectively. By epoch 300, these drops decreased to 17.88% and 9.74%, indicating that the proposed DDSC model maintains higher accuracy throughout training compared to uniform compression.

Regarding training time, both methods showed a similar reduction at epoch 300, with 61.63% for uniform compression and 60.73% for dynamic compression. Although the training time benefits are comparable, the dynamically compressed coordinates provide a significantly smaller loss in translation accuracy. This confirms the advantage of the DDSC model in balancing performance and efficiency. In summary, the DDSC model achieves approximately 60.73% training time reduction with only a 9.74% decrease in accuracy, highlighting its suitability for resource-constrained sign language translation systems.

These findings highlight the practical trade-off between training time and translation accuracy in real-world sign language translation systems. While using all original frames offers the highest accuracy, it requires significantly more computational time and resources. In contrast, the proposed DDSC model achieves a reduction in training time by approximately 60.73% and incurs only a modest decrease in accuracy of 9.74%, demonstrating its viability for resource-constrained environments such as mobile or embedded platforms.

4. Conclusion

This study proposes and evaluates the performance of the proposed DDSC model, which calculates the inter-frame correlation of coordinates extracted from sign language video frames. The proposed DDSC model compresses the coordinates of frames with similar sign language movements as much as possible, while preserving the coordinates of frames with significant movements to prevent any loss of meaning, thereby maintaining the semantic integrity of the sign language.

To further validate the effectiveness of the proposed DDSC model, we compared the translation performance using original, uniformly compressed, and dynamically compressed sign language coordinates under a fixed 75% compression ratio (from 120 frames to 30 frames). The results demonstrate that uniform compression significantly reduces accuracy, with a decrease of approximately 17.88%, due to indiscriminate frame reduction. In contrast, the DDSC model maintains a relatively high level of translation accuracy by selectively preserving semantically meaningful frames. Specifically, the DDSC model achieved only a 9.74% drop in accuracy compared to the original coordinates, while reducing training time by approximately 60.73%. This trade-off highlights the practical advantages of dynamic coordinate compression, especially in resource-constrained environments such as mobile or embedded systems.

In future work, we plan to enhance the DDSC model in several directions. This includes applying it to real-time sign language translation systems, where low-latency and computational efficiency are crucial, enabling deployment on mobile and embedded devices. We also intend to extend the model to sentence-level recognition by addressing the temporal and semantic linkage between individual signs, enabling more natural and fluent translation of continuous sign language. In addition, we aim to incorporate other important features of sign language, such as facial expressions and mouth shapes, to improve the model's ability to capture non-manual signals that are essential for accurate interpretation. We further plan to expand the number of labeled sign language words and perform cross-dataset validation using additional data from AI-Hub and other public resources to improve generalization. Furthermore, due to its coordinate-based approach, DDSC can be adapted to other sign languages such as American Sign Language (ASL) or Japanese Sign Language (JSL), making

cross-linguistic validation an important future direction. In addition, we recognize the importance of comparing performance across different deep learning architectures (e.g., CNN, Transformer, GCN), and this will be addressed in future studies to further validate the generalizability and scalability of the DDSC model.

Acknowledgement

This work was supported by the research grant of Kongju National University Industry-University Cooperation Foundation in 2024, and in part by the MSIT (Ministry of Science and ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation) in 2025 (2024-0-00073).

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